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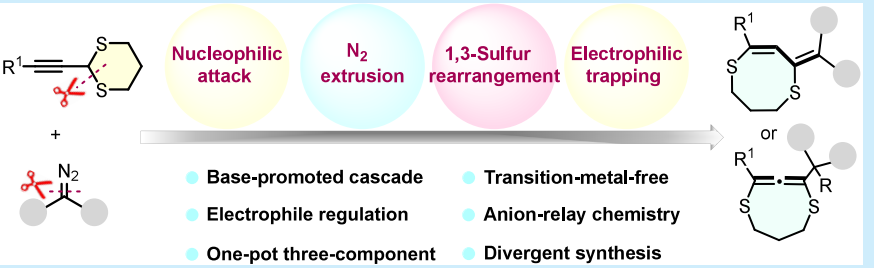
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Access to Conjugated Dienes and Cyclic Allenes via a Diazo-Dithiane Rearrangement Cascade

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ABSTRACT: A transition-metal-free, base-promoted cascade between alkynyl-1,3-dithianes and α-diazodiesters has been developed. The reaction proceeds through an anion relay chemistry (ARC) featuring a selective 1,3-sulfur rearrangement, enabling divergent access to eight-membered sulfur-containing conjugated dienes and cyclic allenes in a one-pot fashion. This transformation unveils new diazo reactivity beyond classical carbene pathways and highlights the synthetic utility of sulfur-based anionic processes.

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Sulfur-containing heterocycles are ubiquitous structural motifs in pharmaceuticals, agrochemicals, natural products, and functional materials.[1] While small sulfur heterocycles such as thiophenes are well established synthetically,[2] the preparation of medium-sized (8−10-membered) sulfur rings remains a formidable challenge owing to transannular strain and conformational rigidity.[3,4] Despite their unique electronic and three-dimensional features, such scaffolds are rarely explored in drug discovery due to the lack of efficient synthetic methods.[5] Therefore, developing general and efficient strategies to access medium-sized sulfur heterocycles under mild conditions is of considerable importance in synthetic and medicinal chemistry.

To date, several strategies have been developed for the construction of medium-sized sulfur-containing heterocycles, including intramolecular transition-metal-catalyzed cyclization, radical-mediated processes, and ring-expansion reactions (Scheme 1a).[6] Among sulfur-rich building blocks, 1,3-dithianes are particularly attractive due to their stability and versatile reactivity beyond their traditional role as carbonyl protecting groups.[7] Early work by Oku demonstrated ring expansion of thioacetals via sulfonium ylide intermediates,[6b] and Terada later achieved an enantioselective variant using chiral phosphoric acid (CPA) catalysis.[6f] Among medium-sized sulfur heterocycles, cyclic allenes and conjugated dienes have attracted particular interest due to their structural uniqueness and potential applications. While significant progress has been made, methods for their general and selective synthesis remain limited. Wang and co-workers reported an Au-catalyzed rearrangement of propargylic 1,3-dithianes to cyclic allenes, with conjugated dienes formed only when specific substituents were present at the 2-position of the dithiane ring.[8a] Similarly, Yucel described a base-mediated rearrangement of 1,3-dithianyl-substituted propargylamines, delivering nine-membered sulfur-containing heterocycles bearing an internal conjugated diene motif under mild conditions.[8b] Although these studies represent elegant contributions, existing strategies remain largely metal-catalyzed and are often limited to specific substrate types. Consequently, the development of transition-metal-free approaches that provide broader synthetic flexibility remains highly desirable.

Diazo compounds have been extensively studied; however, transition-metal-free couplings between diazo species and heteroatom-stabilized carbanions remain underexplored.[9] Metal-free diazo transformations are synthetically appealing because they circumvent potential metal contamination and often proceed under milder conditions. Building on our previous studies of base-promoted anion relay chemistry (ARC) and 1,3-sulfur rearrangements,[10] we envisioned that the nucleophilic anion generated from alkynyl-1,3-dithianes could undergo a carbene-free anionic coupling with α-diazodiesters, followed by a 1,3-sulfur rearrangement to furnish structurally diverse sulfur heterocycles (Scheme 1c). Herein,

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Scheme 1. Background for This Study

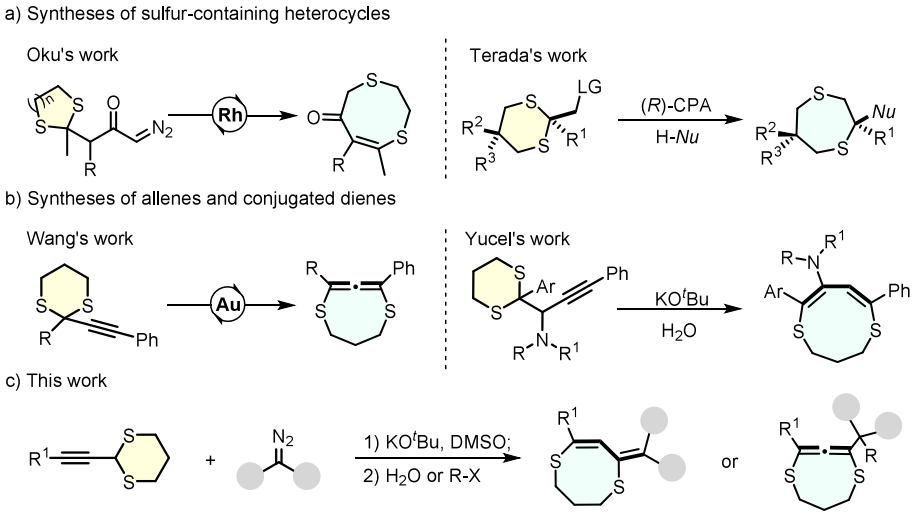
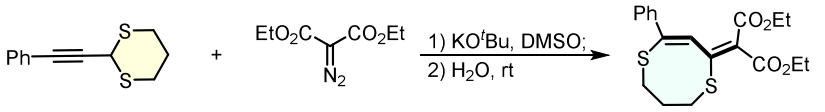


Table 1. Optimization of the Reaction Conditions[a]



1a **2a** **3a**

Entry	Deviation from standard conditions	Yield (%)[b]
1	none	88
2	Without KO^{t}Bu	N.R.
3	NaO^{t}Bu instead of KO^{t}Bu	39
4	DBU instead of KO^{t}Bu	N.R.
5	Cs_2CO_3 instead of KO^{t}Bu	N.R.
6	THF instead of DMSO	N.R.
7	DMF instead of DMSO	65
8	60 °C instead of rt	45

[a]Standard conditions: **1a** (0.1 mmol), **2a** (0.1 mmol), and KO^{t}Bu (0.15 mmol) dissolved in DMSO (2.0 mL), rt, followed by quenching with H_2O to give conjugated diene product **3a**. All entries were quenched with H_2O unless otherwise noted; [b]Isolated yields. N.R. = No Reaction.

we disclose a transition-metal-free, base-promoted cascade reaction between alkynyl-1,3-dithianes and α-diazodiesters that proceeds through an anion relay process. Depending on the electrophilic quencher, this cascade selectively yields eight-membered sulfur-containing conjugated dienes or cyclic allenes, providing a rare example of diazo-dithiane coupling via a concerted anionic rearrangement under mild conditions.

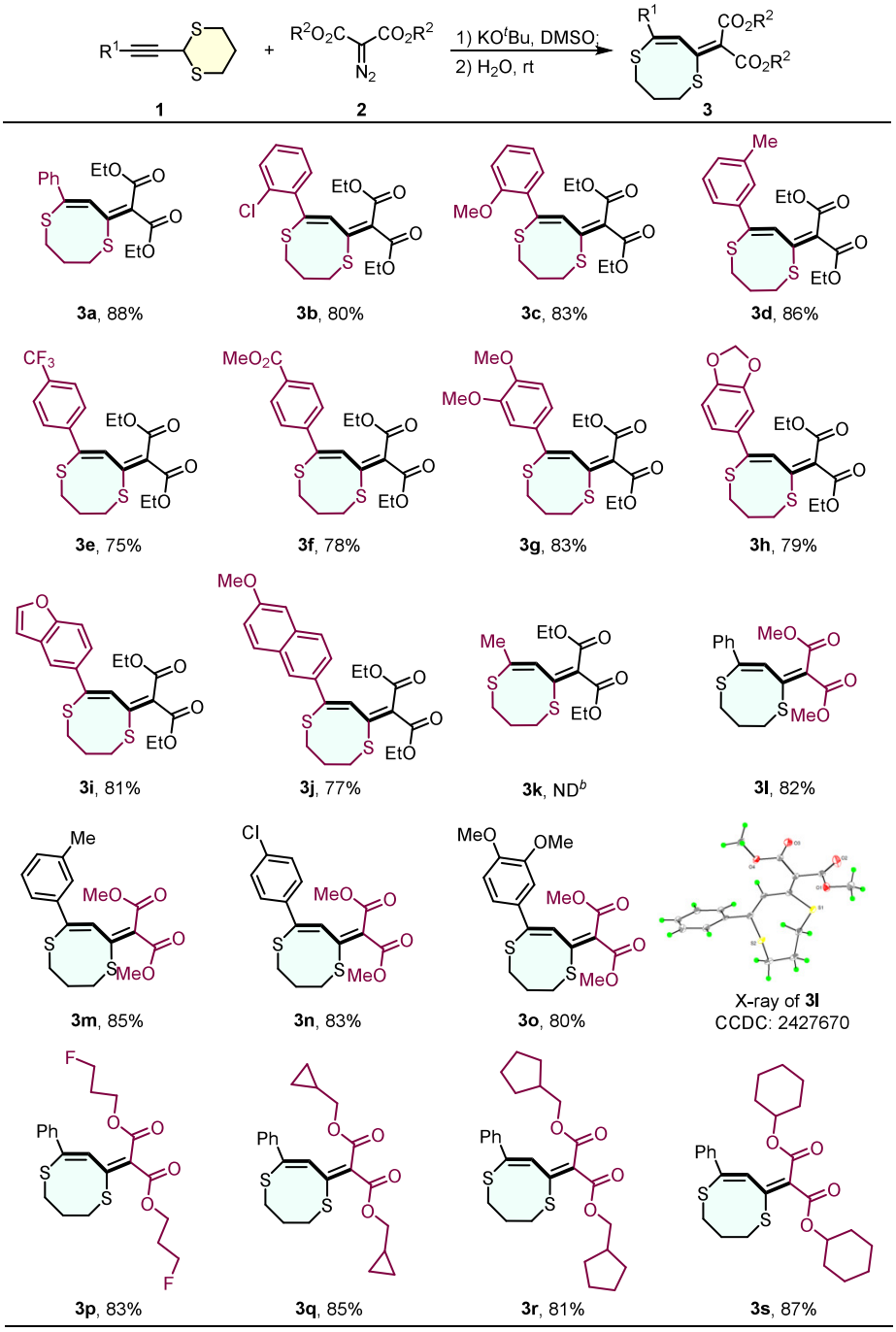
Using alkynyl-1,3-dithiane **1a** and α-diazodiester **2a** as model substrates (Table 1), we identified KO^{t}Bu (1.5 equiv) as the optimal base in DMSO at room temperature, affording the conjugated diene **3a** in 88% yield (Entry 1). Control experiments confirmed the essential role of base and solvent: no conversion was observed without base (Entry 2), weaker bases such as NaO^{t}Bu reduced the yield (39%) (Entry 3), while DBU and Cs_2CO_3 were ineffective (Entries 4 and 5). Among solvents, only DMSO enabled effective conversion; THF completely suppressed reactivity, and DMF led to a moderate yield of 65% (Entries 6 and 7). This unique DMSO effect can be attributed to its ability to generate a nucleophilic DMSO-derived carbanion under basic conditions, thereby initiating the anionic relay, whereas other aprotic solvents fail to provide such activation. Elevated temperature (60 °C) led to decomposition, emphasizing the importance of maintaining mild conditions for this anionic cascade (Entry 8).

With optimized conditions, a wide range of aryl-substituted alkynyl-1,3-dithianes underwent smooth transformation to conjugated dienes **3a**–**3g** in 75–88% yield (Scheme 2). Electron-donating, electron-withdrawing, and sterically hindered substituents were all well-tolerated. Extended aromatic systems (benzofuranyl, piperonyl, naphthyl) also participated smoothly (**3h**–**3j**, 77–81%). In contrast, alkyl-substituted dithianes failed to produce detectable product **3k**, underscoring the stabilizing role of aryl substituents in the intermediate anions. Diverse α-diazodiesters were also compatible, furnishing products **3l**–**3s** in 80–87% yield. The structure of **3l** was unambiguously confirmed by X-ray crystallography (CCDC 2427670).

Remarkably, the reaction outcome could be significantly modulated by the choice of quenching reagent. Replacement of H_2O with electrophilic alkyl halides (e.g., methyl iodide, benzyl bromide, ethyl bromoacetate, 1-iodononane, and iodocyclopentane) led to sulfur-containing cyclic allenes **4a**–**4i** in 71–86% yield (Scheme 3). This switchable reactivity underscores the tunability of the anionic relay pathway and offers a concise, operationally simple approach to cyclic allene frameworks, which serve as valuable intermediates for complex molecule synthesis.

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Scheme 2. Substrate Scope of Conjugated Dienes



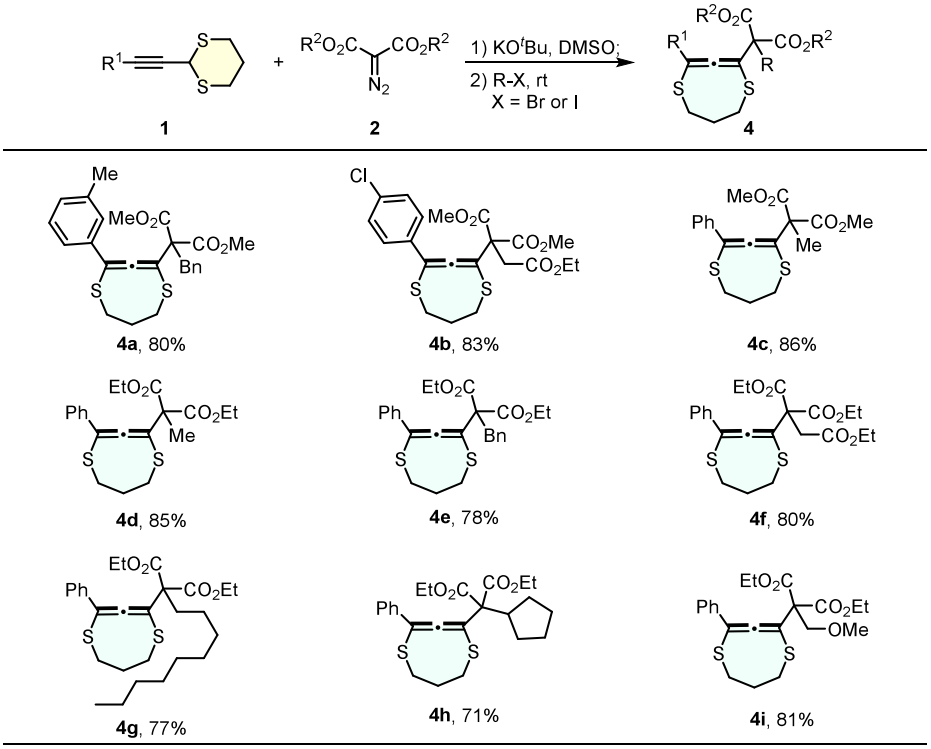
[a]Reaction conditions: **1** (0.1 mmol), **2** (0.1 mmol), KO^{t}Bu (0.15 mmol) in DMSO (2.0 mL), rt, then quenched with H_2O. Isolated yields. [b]ND = Not Detected.

The transformation was readily scaled up to gram scale, affording **3a** in 80% yield (Scheme 4a). The conjugated diene product was further diversified through selective functionalization (Scheme 4b). Reduction of the ester group using DIBAL-H at different temperatures afforded the corresponding primary alcohol **5a** in 56% yield or the aldehyde **5b** in 53% yield. Furthermore, treatment of **3a** with Raney-Ni under an air atmosphere led to efficient desulfurization and hydrogenation, yielding the saturated compound **5c** in 72% yield. In addition, oxidation of the dithiane unit with *m*-CPBA enabled a mild and selective conversion to the sulfoxide derivative **5d** in 78% yield.

Radical inhibition experiments with TEMPO showed no effect, ruling out radical pathways.[8b] The reaction of 97% deuterated-1,3-dithiane **D-1a** with α-diazodiesters **2a** gave product **D-3a1** with 9% D incorporation at the vinylic position. When the crude reaction mixture was quenched with D_2O, product **D-3a2** was obtained with 96% D incorporation at the same position. In contrast, quenching with DMSO-d_6 afforded **D-3a3** with only 6% D incorporation. Deuterium-labeling studies revealed that deprotonation of 1,3-dithiane at the α-

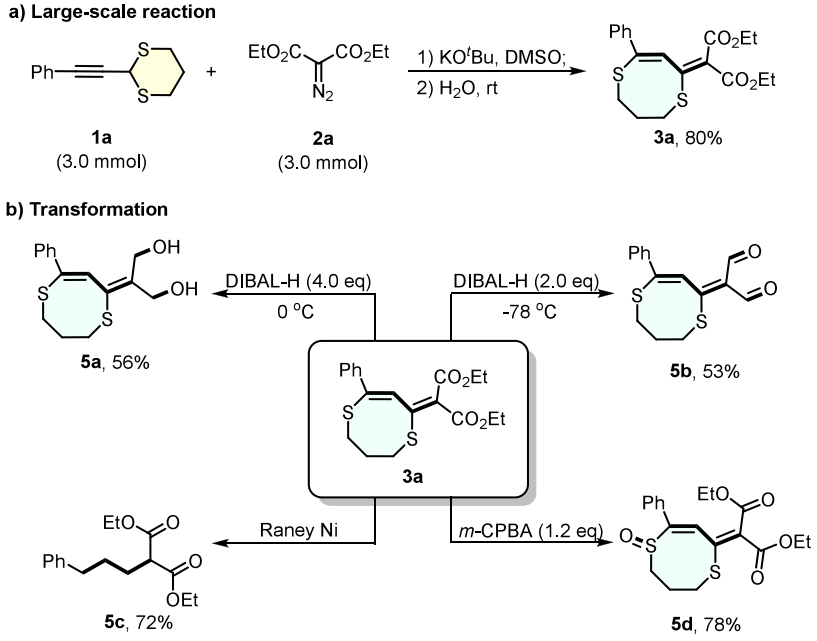
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Scheme 3. Substrate Scope of Cyclic Allenes[a]



[a]Reaction conditions: **1** (0.1 mmol), **2** (0.1 mmol), KO^{t}Bu (0.15 mmol) in DMSO (2.0 mL), rt, then quenched with electrophiles (0.15 mmol) to trap the reactive intermediate. Isolated yields.

Scheme 4. Scale-up Experiments and Synthetic Utility

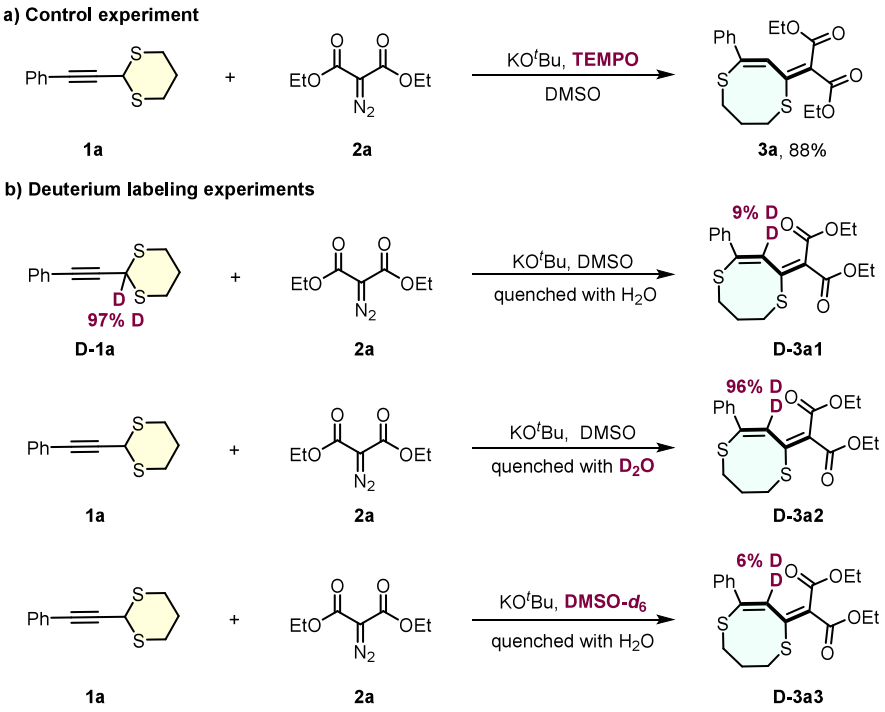


position precedes coupling, and the protonation occurs exclusively during the quenching step. These results support an anionic mechanism involving sequential deprotonation, cyclization, and sulfur migration (Scheme 5).

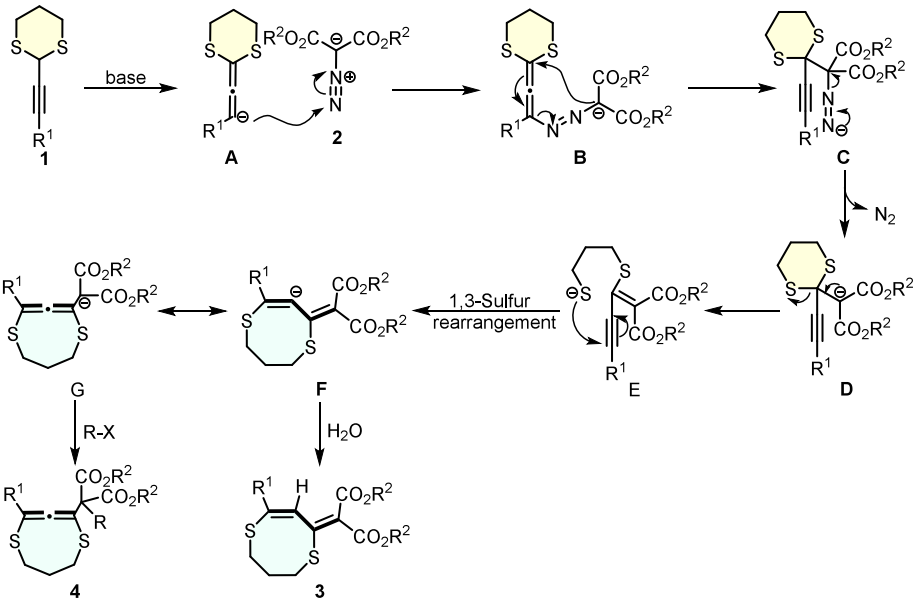
A plausible mechanism consistent with the experimental observations is depicted in Scheme 6. Deprotonation of alkynyl-1,3-dithiane **1** generates an alkynyl anion, which rapidly equilibrates to the allenyl anion **A**. Nucleophilic attack of **A** to the α-diazodiester **2** furnishes intermediate **B**, followed by an intramolecular nucleophilic attack to afford **C**. Subsequent extrusion of N_2 forms anion **D**, which undergoes an anion-relay process, triggering fragmentation of the 1,3-dithiane scaffold to generate a sulfur-centered anion **E**. A 1,3-sulfur rearrangement, driven by the reorganization of the C−S bonds, converts **E** into the conjugated diene anion **F**. Protonation of **F** during aqueous workup furnishes the

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Scheme 5. Mechanism Investigation



Scheme 6. Proposed Mechanism



observed diene **3**. Alternatively, anion **F** can tautomerize to the allenyl anion **G,** whose interception by an external electrophile leads to the formation of the cyclic allene **4**.

In summary, we have developed a transition-metal-free, anion-relay cascade reaction between alkynyl-1,3-dithianes and α-diazodiesters that enables the divergent synthesis of conjugated dienes and cyclic allenes. This transformation features a distinctive 1,3-sulfur rearrangement under mild basic conditions, exhibits broad substrate scope with tunable selectivity controlled by the electrophilic quencher, and proceeds with operational simplicity and scalability. Importantly, this work expands diazo chemistry beyond conventional carbene reactivity into the realm of anionic processes, demonstrating the potential of sulfur-based anionic cascades for the efficient construction of medium-sized heterocycles and offering a new conceptual framework for transition-metal-free diazo coupling chemistry.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c04254.

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Experimental details, characterization data, X-ray crystallographic data of **3l**, and copies of ^{1}H and ^{13}C NMR spectra (PDF)

Accession Codes

Deposition Number 2427670 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

Z.Z. and X.G. contributed equally to this work. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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